

JC4004 – Computational Intelligence

Sessions 1 & 2: Introduction

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Icebreaker activities

- Who am I?
- How familiar are you with artificial intelligence?
- What do you expect from this course?

Course arrangements

- **Lectures** (theory classes) 2 times / 5 sessions a week (48 sessions in total)
- **Practical assignment**, 30% of the assessment
 - Will be released 22 November, more details later
 - Feedback no longer than three weeks after the deadline
- **Exam**, 70% of the assessment

Course outline

- Week 1: Introduction, Game theory, Reinforcement Learning, fuzzy logic, genetic algorithms
- Week 2: Neural networks, deep learning basics
- Week 3: Advanced neural networks, attention and transformers
- Week 4: Multimodal and generative AI, future trends

Academic integrity

- Academic integrity is expected in the coursework assignment!
Examples of violation of academic integrity:
 - **Plagiarism**: the unattributed use of other people's work or ideas, in work submitted for assessment
 - **Collusion**: unauthorised collaboration between students in the writing of an individual assignment
 - **"Contract cheating"**: a form of academic misconduct where a student submits work that has been produced by someone other than the student, whether this has been paid for or not
- Link: <https://www.abdn.ac.uk/students/academic-life/academic-integrity.php>

Plagiarism in reports

- If taking text verbatim, put it in quotes and *cite the source*
- Alternatively, paraphrase or summarize and *cite the source*
- As a rule of thumb, don't do this for more than 3-4 sentences
- See <https://integrity.mit.edu/handbook/writing-original-work> for an excellent guide

Note that using large language models (e.g., ChatGPT) to write your reports or code can also be considered as plagiarism!

Plagiarism in source code

- An excellent guide: <https://integrity.mit.edu/handbook/writing-code>
 - Never copy another student's code
 - Don't simply reuse code directly
- Cite the sources of code that you use within your code
- Ensure that any licensing conditions are adhered to

Note that using large language models (e.g., ChatGPT) to write your reports or code can also be considered as plagiarism!

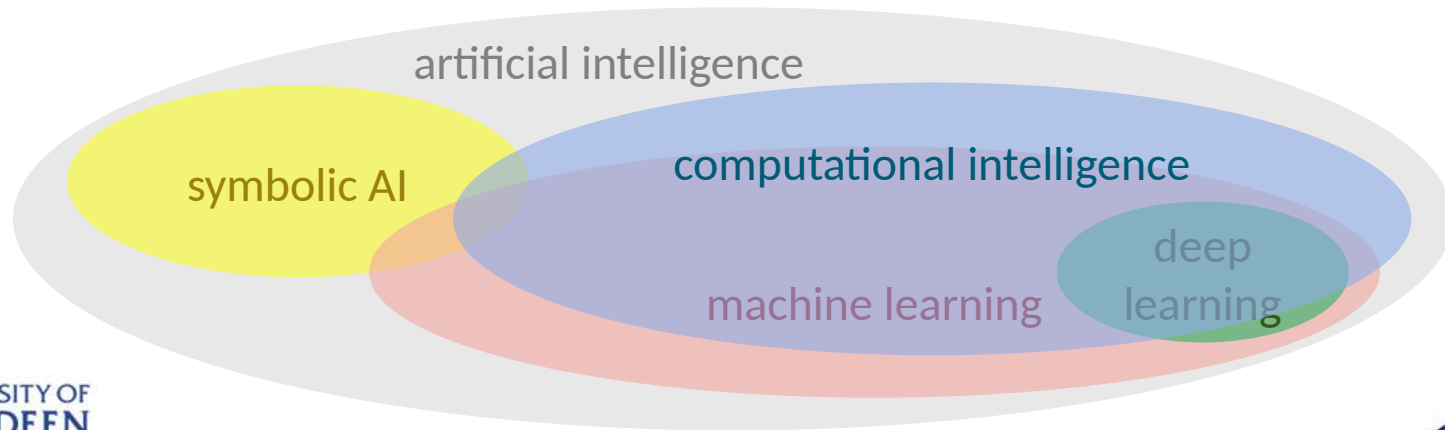
Ask for help if you are stuck

- If something is unclear, then ask someone or look it up
- If you still find it hard, get in touch with your teacher
- Copying something from someone else, or via the internet, will not help you learn how to do the work

It is ALWAYS better to submit a poor assignment of your work with comments about what you tried, and struggle with, than to submit someone else's work.

What is computational intelligence?

- *Computational intelligence (CI)* usually refers to the ability of a computer to learn a specific task from training data
 - Subfield of *artificial intelligence (AI)*, focused on development of biologically and linguistically motivated AI systems
 - However, terminology is not always used consistently...



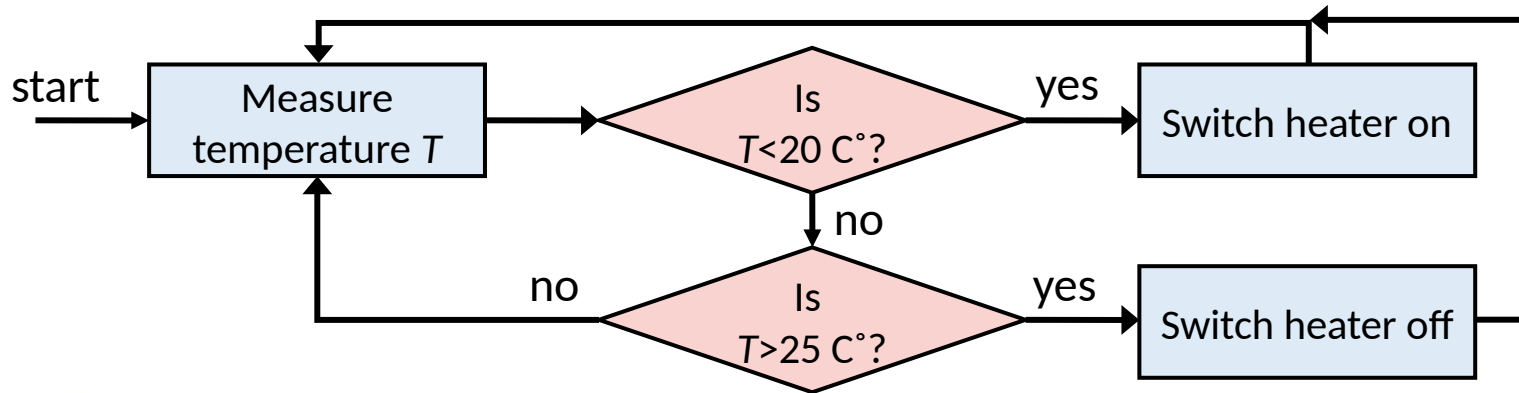
Artificial general intelligence



- Early philosophers and AI researchers envisioned AI to achieve *artificial general intelligence (AGI)*
 - Machines matching or surpassing human capabilities across a wide range of cognitive tasks
 - AGI turned out to be difficult to achieve: to date, AI has been mostly successful just in narrowly specified tasks (*narrow AI*)
 - However, recent large language models seem to approach AGI
- Related to AGI, we can define *strong AI* and *weak AI*
 - *Strong AI* is self-conscious AI
 - *Weak AI* is AI that only *acts* like it was self-conscious

Symbolic AI

- *Symbolic AI* (classical AI) refers to methods based on *symbolic* (human readable) representations and reasoning to perform tasks
 - *Good old-fashioned AI* (GOFAI), based on hand-written rules
 - Dominating AI paradigm from 1950s to mid-1990s



Pros and cons of symbolic AI

Advantages

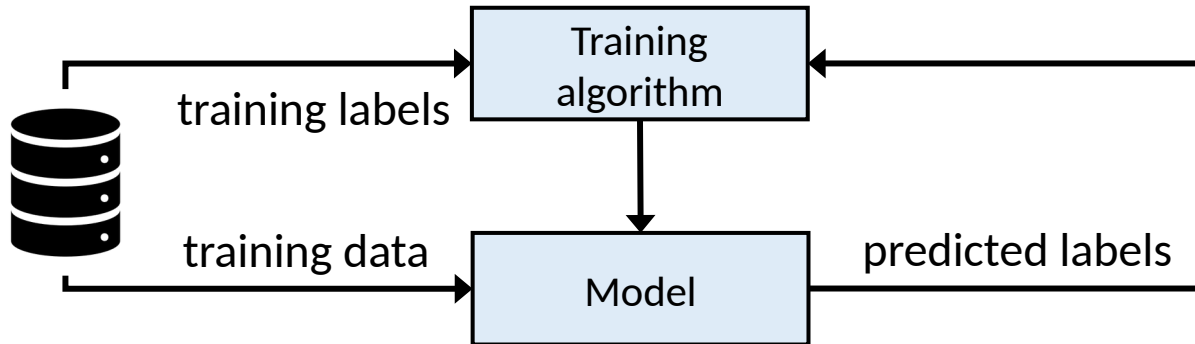
- Interpretability: enables transparent and explainable reasoning
- Rule-based approach: easy to modify and debug
- Low complexity: can be often implemented with limited computational resources

Disadvantages

- Knowledge acquisition: requires a lot of effort and expert knowledge to implement the rules
- Poor scalability: difficulties to handle large, unstructured datasets
- Poor adaptability: struggles to cope with uncertainties, new or incomplete information

Non-symbolic AI

- *Non-symbolic (or connectionist AI)* refers to methods where the rules are not hand-written but learned using training data and related labels
 - Training algorithm adjusts the internal model parameters (e.g., neural network weights) to generate accurate predictions
 - Dominating AI paradigm since the mid-1990s



Pros and cons of non-symbolic AI

Advantages

- Adaptability: adapts to unseen data and patterns
- Scalability: can handle large and complex datasets with undefined input data type
- Robustness to noise: makes predictions even on noisy or imperfect data

Disadvantages

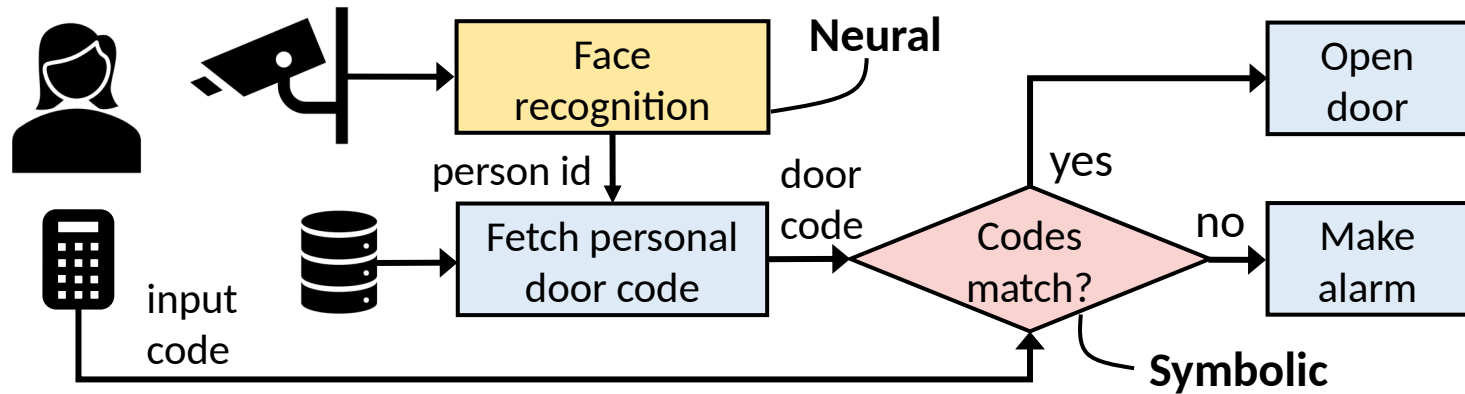
- Poor interpretability: trained model is a “black box”: decision-making process difficult for humans to understand
- Data dependency: depends on large high quality training datasets
- High complexity: often requires a lot of computational resources for training

Symbolic versus non-symbolic AI

- Choice of symbolic or non-symbolic AI depends on the application
 - Symbolic AI requires expert knowledge but can be used without training data, whereas non-symbolic AI requires high quality training data
 - Symbolic AI excels in well-defined tasks with specific data, whereas non-symbolic AI copes better with uncertainties and unstructured data
- In this course, we mostly deal with non-symbolic AI
 - However, distinction between symbolic and non-symbolic AI is not always clear: e.g., fuzzy logic can rely on either hand-written or learned rules

Neuro-symbolic AI

- In many applications, non-symbolic neural networks are used together with symbolic reasoning
 - There are several different ways to combine neural networks with symbolic AI into *neuro-symbolic AI*

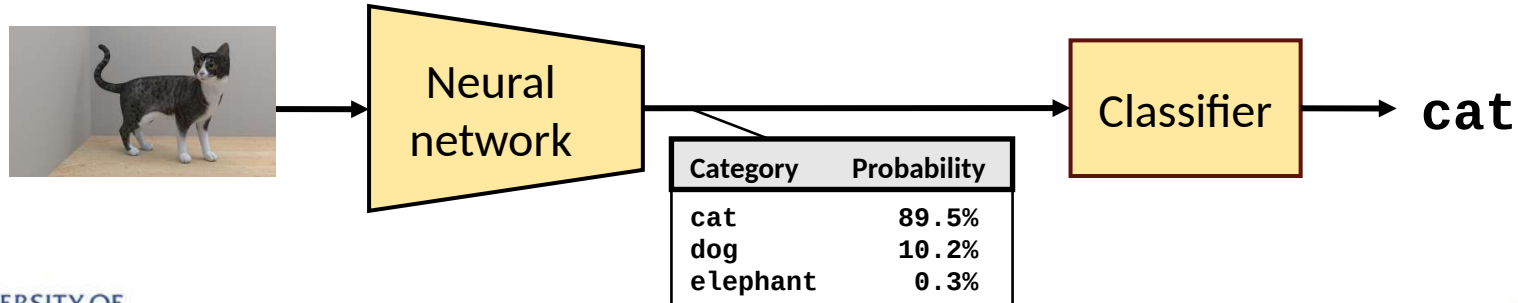


Moravec's paradox

- Early AI researchers assumed that human cognitive functions could be reduced to precise logical rules, as in symbolic AI
- However, **Moravec's paradox** was later observed: *easy tasks to humans are difficult to AI, and difficult tasks to humans are easy to AI*
 - Tasks such as face recognition require a lot of computations, but they are easy to humans, because our brains are evolved to do those tasks
 - Tasks such as solving mathematical equations or playing chess do not require as many computations, but they are difficult to humans, because our brains are not evolved to do that kind of tasks

Soft computing

- Soft computing refers to algorithms giving approximate solutions to problems that are impossible or too costly to be solved accurately
 - Many problems covered by Moravec's paradox that are difficult to AI but easy to humans fall within this category
 - Many soft computing methods are also non-symbolic AI methods
 - Examples: *fuzzy logic, evolutionary algorithms, neural networks*



Break: any questions?

History of CI: precursors



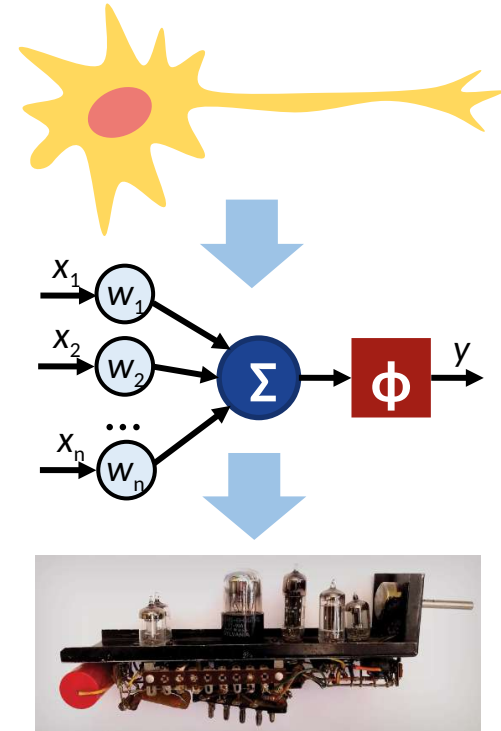
René Descartes

1596-1650

- The idea of “artificial man” capable to think was known in Greek mythology and medieval legends already
- Ancient Chinese, Indian, and Greek philosophers developed methods for formal reasoning
- Different mechanical calculating devices built throughout history by several people
- In the 17th century, philosophers such as **Leibniz**, **Descartes**, and **Hobbes** speculated that human reasoning could be reduced to mechanical calculation

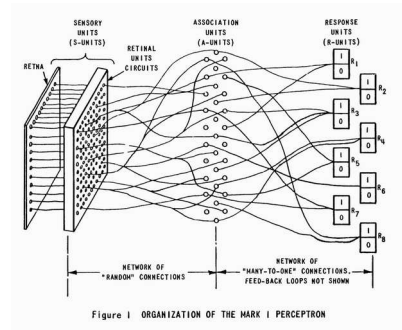
History of CI: the birth of AI

- Advances in formal logic, electronic computers, and neurosciences paved way for modern AI in the early 20th century
- Symbolic reasoning algorithms implemented in electronic computers in 1950s
- **Walter Pitts** and **Warren McCulloch** theorised artificial neurons and neural networks in 1940s
- **Marvin Minsky** and **Dean Edmonds** built the first neural net machine *SNARC* in 1951



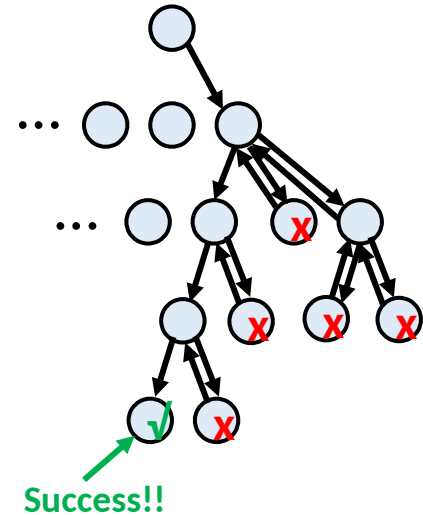
History of CI: perceptrons

- *Perceptron* is a neural network for binary classification
 - Generates a vector of numbers to decide if the input belongs to a specific class
 - Invented by Pitts and McCulloch in 1943
- The first hardware implementation by **Frank Rosenblatt** in 1957
 - Three layers: input layer with 20 x 20 photocells, hidden layer of 512 neurons, and output layer of 8 neurons
 - Weights adjusted with electric motors



History of CI: early success

- From 1950s to 1960s, AI was able to solve many problems successfully
 - Lot of enthusiasm and funding available
 - Symbolic AI was the dominating paradigm, but some work also on perceptrons and neural nets
- Many practical problems were solved with simple search algorithms
 - All possible paths attempted, returns if there is a dead end, continues until solution found
 - Not suitable method for large problems!

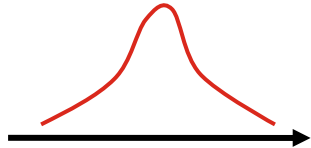
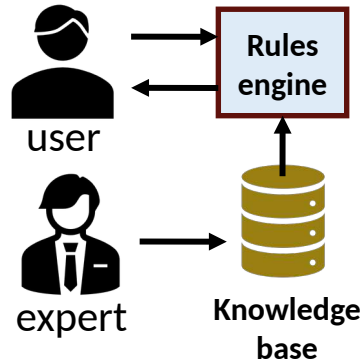


History of CI: the first AI winter



- After the early success, progress in AI slowed down
 - Hardware limitations (memory and computational power)
 - Early algorithms (such as search trees) were not suitable for large problems, due to combinatorial explosion
 - *Moravec's paradox*: what is trivial to human is often difficult to AI
 - Many real-world problems and concepts are imprecise by nature, and cannot be solved using formal logic
- Disappointing results reduced funding and led to the first *AI winter* from 1960s to early 1980s

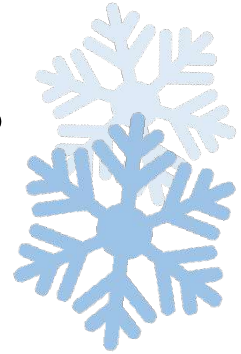
History of CI: the boom in the 1980s



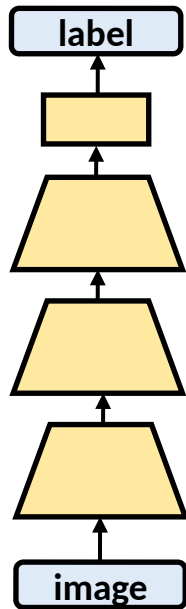
- Adoption of “expert systems” using symbolic AI in a restricted domain of knowledge in many applications
- Soft computing and probabilistic reasoning popularised
 - Useful for solving imprecisely defined problems
 - Techniques like fuzzy logic (proposed by **Lofti Zadeh** in 1960s) and evolutionary computing (originating from 1950s) were adopted in practical applications in 1980s
- *Backpropagation* first used for training neural networks by **David Rumelhart**, **Geoffrey Hinton**, and **Ronald Williams** in 1986

History of CI: the second AI winter

- Many AI companies fell in the economic bubble in the 1980s
 - Funding to AI projects declined again in late 80s and early 90s
- However, a lot of AI research still going on “behind the scenes”
 - Improved computer hardware technology allowed further development of techniques such as neural networks
 - Due to the poor reputation of AI, researchers adopted different terms such as “cognitive systems” or “computational intelligence”
 - Focus on solving specific practical problems, i.e. *narrow AI*, rather than creating *artificial general intelligence*

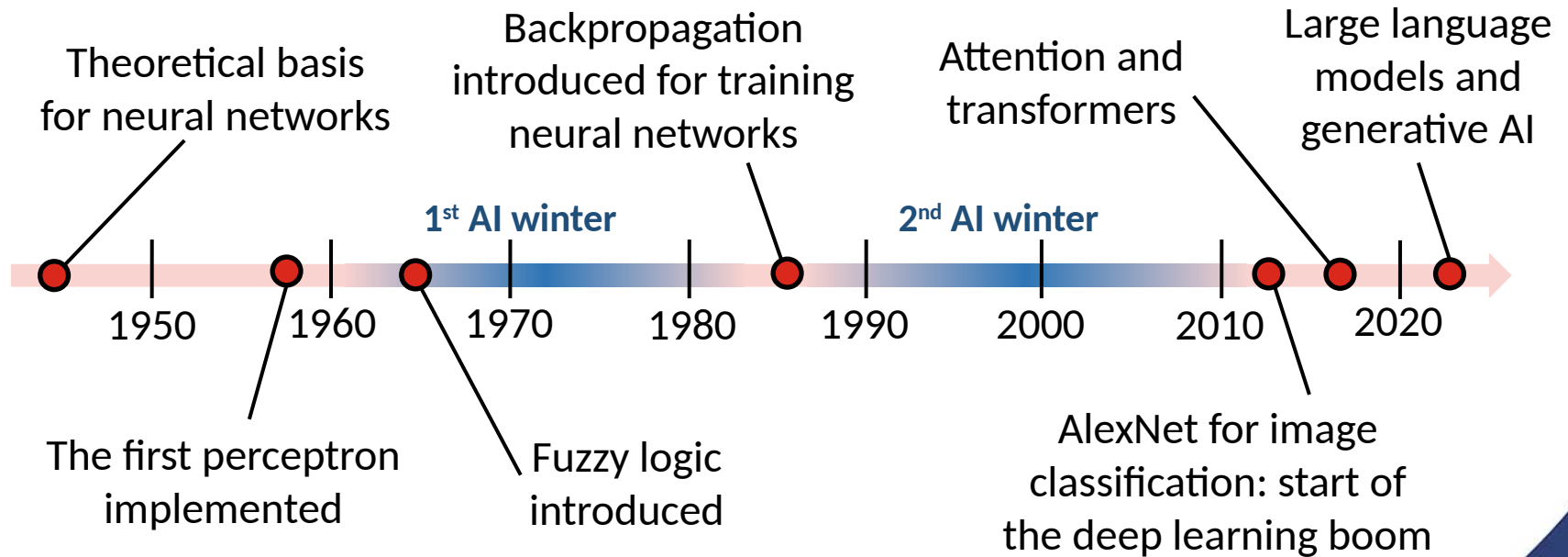


History of CI: deep learning



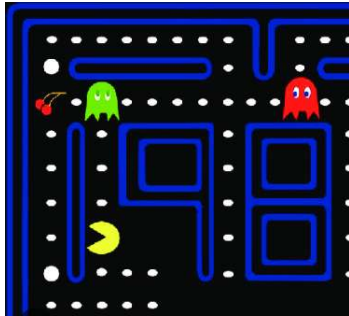
- More powerful computers allowed development of deep neural networks and handling of big data since mid-2000s
 - Deep learning uses neural networks with a lot of layers, requiring a lot of computational power and training data
 - Deep learning has achieved remarkable success in challenging tasks like image and speech recognition
- In the past few years, deep AI models able to generate text seemingly written by humans, as well as natural-looking images, have been introduced
 - More about recent developments later in this course...

Summary of main milestones in AI



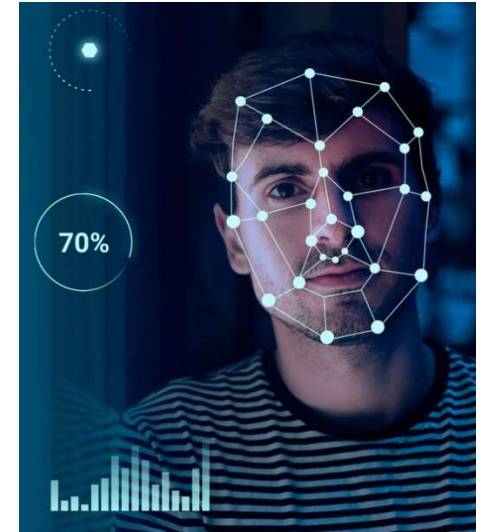
Applications: video games

- In video games, AI is used to control the actions of the non-playable characters (NPC) to model human-like behaviour
 - In the early games, simple symbolic rules were used
 - In the modern games, non-symbolic AI such as neural networks are applied to achieve more realistic NPC actions



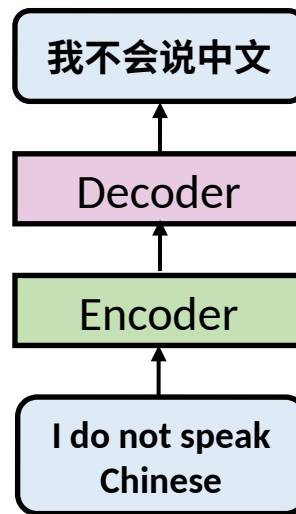
Applications: face recognition

- Face recognition is one of the most essential problems that are easy to humans but hard for AI
 - Several practical applications, such as access control and identity verification at border stations etc.
 - Early traditional methods tried to identify positions of facial landmarks (cheek, eyes, nose), but they were not robust to noise like shadows
 - The recent methods based on deep learning have achieved sufficient accuracy for practical applications



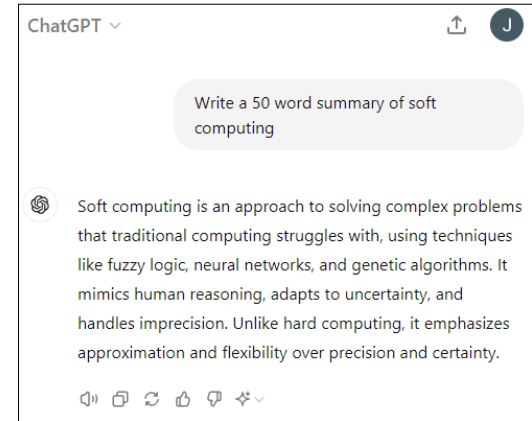
Applications: machine translation

- Machine translation can greatly help communication when people speak different languages
 - Translation word-by-word does not work well, because meaning of many words depends on the context
 - Encoder-decoder model architectures first encode input in the internal representation, then decode it in the target language
 - Transformer-based encoder-decoder systems take the context into account and generate accurate translations



Applications: text generation

- Recently developed large language models, such as ChatGPT, are capable to write short texts or essays in flawless language
 - Trained using a massive amount of text corpora from various sources
 - Several practical applications, such as chatbots or writing summaries of data
 - Shows capability of human-like reasoning

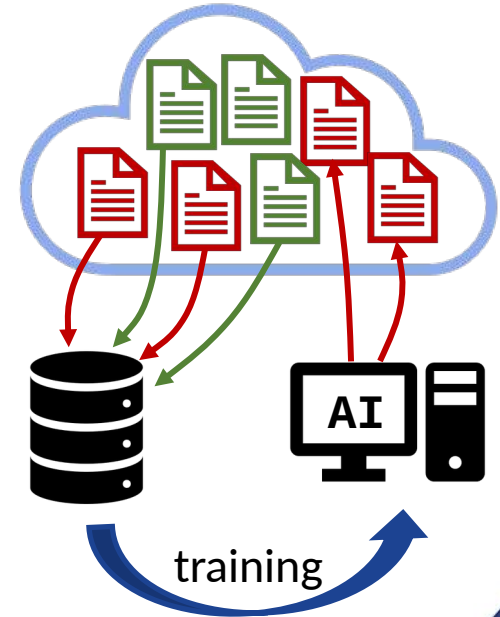


Ethical and legal considerations

- Like any technology, CI can be used for good and bad purposes
 - Realistic images or speech generated by AI (i.e. *deepfakes*) can be used to deceive and cheat people
- Quality of deep AI systems depends on the quality of training data
 - Biases and errors in training data cause biases and errors in the system
- Legislation and regulations are not fully prepared for the AI era
 - Who owns the copyrights of the works created by AI?
 - Who is responsible for damages caused by AI systems?
 - Generative AI and academic integrity?

Model collapse

- More and more AI-generated content is published on the Internet: more and more AI-generated content will also be used for training AI models!
- When AI-generated content is used recursively to train AI models, the models will degenerate and eventually collapse
- The problem of model collapse is not (yet) urgent, but it must be taken into consideration!



Summary

- Course practicalities introduced
 - 48 theory sessions (4 times per week, 2 or 4 sessions each time)
 - Assessment: 70% exam, 30% coursework assignment
- This course focuses mostly on non-symbolic AI techniques
 - Fuzzy logic, evolutionary computing, neural networks
 - Focus on deep neural networks and the advances within the past 20 years
- Basic concepts and brief history of CI introduced
 - Symbolic vs. non-symbolic AI, Moravec's paradox, soft computing
 - Example applications: video games, face recognition, etc.

Questions and discussion